

Proposal for a Verified Dating Platform

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Application Type: **Cross-Platform Mobile Application**

1. Executive Summary

Online dating platforms have significantly changed how people connect globally. However, modern dating applications face serious challenges such as **catfishing, identity fraud, privacy concerns, and lack of trust between users**.

This proposal outlines the development of a **cross-platform dating application focused on identity verification, user safety, and privacy-first communication**.

The primary goal of the platform is to create a **trust-driven dating ecosystem** where users can evaluate the credibility of profiles through **tiered verification levels**, while maintaining strong **privacy controls and secure communication**.

The platform will be built using **React Native for cross-platform mobile development** and a **Node.js backend architecture**, ensuring scalability and performance for a global audience.

2. Problem Statement

Modern dating platforms face several persistent issues that reduce user trust and safety.

2.1 Catfishing & Fake Profiles

Many users create fake profiles using stolen images or misleading information to deceive others.

This results in:

- Emotional harm
 - Scams and fraud
 - Reduced trust in the platform
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2.2 Privacy Risks

Users are often exposed to privacy risks such as:

- Screenshots of private conversations
 - Unwanted sharing of personal information
 - Early exposure of sensitive data like phone numbers or addresses
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2.3 Inefficient Matching Systems

Many applications rely on simplistic matching algorithms that do not consider important user boundaries.

Users often want:

- Non-negotiable preferences
- Lifestyle compatibility
- Cultural or religious considerations

Without proper filtering, user experience deteriorates.

3. Proposed Solution

We propose a **Verified Dating Platform** designed to prioritize **trust, privacy, and compatibility**.

The platform introduces the following key innovations:

1. Incentivized Verification System

Users can choose multiple levels of verification to increase their **trust score**.

2. Privacy-First Communication

Strict privacy controls prevent unauthorized sharing of sensitive information.

3. Intelligent Matching Filters

A hybrid filtering system using **Hard Filters and Soft Filters** ensures compatibility while maintaining inclusivity.

4. Core Features

4.1 Multi-Level Profile Verification

To eliminate catfishing, the platform introduces a **three-tier verification model**.

Level 1 — Basic Verification

- Face recognition during onboarding
- Live selfie validation
- Ensures the user matches uploaded photos

Level 2 — Trustworthy Verification (Optional)

Users can connect verified social accounts such as:

- Instagram
- Facebook
- Snapchat

Additionally:

- Government ID verification may be performed.

Level 3 — Secure Verification (Optional Future Feature)

A deeper trust level may include:

- Background verification
- Criminal record checks (subject to legal and regional compliance)

This tier is optional to maintain **low onboarding friction while enabling higher trust levels for serious users.**

4.2 Privacy Protection System

User privacy will be protected through multiple mechanisms.

Screenshot Protection

- Android: Screenshot blocking using secure window flags
- iOS: Screenshot detection with warning and potential content masking

Personal Data Protection

Sensitive information such as:

- Phone numbers
- Addresses

will remain hidden until **both users mutually approve sharing after matching.**

Reporting System

Users can report profiles for:

- harassment
- inappropriate content
- suspicious behaviour

4.3 Matching System

The application will implement a **hybrid matching model**.

Hard Filters (Non-Negotiable)

These filters strictly control which profiles are shown.

Examples include:

- Orientation
- Religion
- Relationship goals
- Age range

Soft Filters

These influence but do not strictly limit matching.

Examples include:

- Distance radius
- Lifestyle preferences
- Dating intentions (casual / serious / unsure)

This approach ensures **inclusivity while respecting user boundaries**.

5. Messaging Features

The platform will include modern messaging features similar to those present in leading dating applications.

Core messaging capabilities include:

- Text messaging
- Voice messages

- Image sharing
- Video sharing
- Read receipts
- Typing indicators
- Disappearing messages
- Media moderation
- Optional video calling (future phase)

Messaging will only unlock **after mutual matching** between users.

6. Location Based Matching

The application will support **radius-based matching**.

Users can define a preferred distance range (for example 5km–100km).

Location will be used only to:

- show nearby profiles
- improve matching relevance

Exact locations will **never be shared publicly**.

7. Safety Features

Safety is a key differentiator of the platform.

The application will include:

User Reporting

Instant reporting of suspicious or abusive accounts.

Profile Flagging System

Repeated reports will automatically flag profiles for review.

Emergency Contact Feature (Future Phase)

Users may optionally add emergency contacts when going on a date.

Date Safety Check-ins (Future Phase)

Users can notify trusted contacts when meeting someone.

8. Admin Dashboard

An administrative dashboard will allow platform moderators to manage the system effectively.

Key admin capabilities include:

User Management

- Ban users
- Suspend accounts
- Delete accounts

Verification Management

- Review verification submissions
- Approve or reject verification levels

Content Moderation

- Review reported messages
- Review reported profiles

Analytics

- User activity
- Match success rate
- Safety reports

9. Technical Architecture (Subject to changes)

The platform will follow a **modern scalable architecture**.

Mobile Application

Built using:

- React Native

This allows a single codebase for:

- iOS
 - Android
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Backend Server

The backend will be built using:

- Node.js
- Express.js

Responsibilities include:

- authentication
 - matchmaking logic
 - messaging services
 - verification services
 - moderation systems
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Database Layer

Primary database:

- PostgreSQL

For:

- user profiles
- matches
- chat metadata
- verification data

Real-Time Messaging

Real-time communication will use:

- WebSockets (Socket.IO)

For:

- instant chat
 - typing indicators
 - message delivery
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Caching Layer

Redis will be used for:

- session storage
 - caching matches
 - rate limiting
 - real-time data optimization
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10. Third-Party Services

Several external services may be required.

Authentication

- Firebase Auth
- Auth0

Face Verification

- AWS Rekognition
- FaceIO
- Onfido

Identity Verification

- Onfido
- Persona
- Sumsub

Cloud Storage

- AWS S3
- Cloudflare R2

Used for:

- profile images
- media uploads

Push Notifications

- Firebase Cloud Messaging
- OneSignal

AI Moderation (Future Integration)

- AWS Rekognition
- Google Vision API

Used for:

- nudity detection
- spam detection
- harassment monitoring

Maps & Location

- Google Maps API

11. Security & Compliance

The platform will implement strict security measures.

Data Encryption

All sensitive data will be encrypted using:

- HTTPS / TLS
- secure token authentication

Authentication

- JWT based authentication
- refresh tokens

Rate Limiting

Protection against bot attacks and spam.

GDPR Considerations

Since the application targets a global audience, the system should follow:

- GDPR privacy standards
 - user data deletion requests
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14. Conclusion

This proposed platform addresses major issues in online dating through a **verification-first ecosystem**, **privacy-focused design**, and **secure communication infrastructure**.

By prioritizing **user trust and safety**, the application has the potential to stand out significantly in the modern dating app landscape.

The proposed architecture ensures that the platform can scale globally while maintaining performance, security, and reliability.

15. Total Estimated Development Duration

Approximately 12 – 16 Weeks (2 – 3 Months)

This timeline may vary depending on:

- feature changes
- AI integrations
- verification integrations
- additional messaging features